# How SVM Kernels Shape Decision Boundaries: Linear, Polynomial, and RBF on Toy Datasets

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*Course: Machine Learning Tutorial Assignment*

***Abstract— Support Vector Machines (SVMs) are a popular supervised learning method for classification, but their behaviour depends strongly on the choice of kernel function[1][2]. In this tutorial, I explore how three standard kernels—linear, polynomial, and radial basis function (RBF)—change the decision boundary learned by an SVM on simple two-dimensional datasets. Using scikit-learn's synthetic make_blobs, make_moons, and make_circles datasets, I train SVM classifiers with each kernel and visualise the resulting decision regions[3][4]. The experiments show that linear kernels work well when classes are linearly separable, while polynomial and RBF kernels are better suited to curved or nested patterns[3][5][6]. I summarise the performance of each kernel, provide practical guidelines for kernel choice, and briefly discuss ethical considerations around overfitting and responsible deployment of SVM-based models.***

## I. Introduction: SVMs and Kernels

Support Vector Machines (SVMs) are margin-based classifiers that search for a decision boundary which maximizes the distance between the closest points of each class, known as support vectors [1][2][7]. In the simplest case, an SVM with a linear kernel finds a straight line (or hyperplane) that separates two classes in the original feature space [3][8]. When the data are linearly separable, this simple model can perform very well.

Many real-world datasets, however, are not linearly separable: the classes may interleave, form concentric shapes, or follow more complex manifolds [6]. To handle such cases, SVMs use kernel functions to implicitly map the data into a higher-dimensional feature space where a linear separator may exist [6][9]. Common choices include the linear kernel, the polynomial kernel, and the radial basis function (RBF) kernel [1][5].

This tutorial focuses on building intuition for how different kernels shape the decision boundary in simple two-dimensional problems. Rather than deriving all the mathematics, I use visual experiments to answer three questions:

- When does a linear kernel work well?
- How do polynomial and RBF kernels change the shape of the boundary?
- How should we think about kernel choice in practice?

All experiments are implemented in Python using scikit-learn's SVC class for SVM classification and standard toy datasets [1][3][4]. The accompanying Jupiter notebook reproduces all figures and results and is available in the GitHub repository linked at the end of this document.

## II. Experimental Setup

All experiments use synthetic two-dimensional datasets generated with scikit-learn:

- **Blobs:** Two Gaussian clusters created with make_blobs, which are almost linearly separable[3].
- **Moons:** Two interleaving half moons from make_moons, a standard non-linear classification benchmark[4].
- **Circles:** Two concentric circles from make_circles, representing nested non-linear structure[3][6].

For each dataset I sample 200 points, then split them into 70% training and 30% test sets using a fixed random seed (RANDOM_STATE = 42) for reproducibility. I train SVM classifiers with the following kernels:

- **Linear:** kernel="linear", $C = 1.0$.
- **Polynomial:** kernel="poly", degree $d = 3$, gamma="scale", $C = 1.0$.
- **RBF:** kernel="rbf", gamma="scale", $C = 1.0$ [1][5][6].

Models are implemented using scikit-learn's SVC class[1]. For each model, I report test accuracy and visualise the decision boundary by predicting class labels on a dense grid over the

feature space. Plots use a colour-blind-friendly palette (blue and orange) and different marker shapes (circles and triangles) for each class to improve accessibility[10].

The goal is not to optimize hyperparameters but to compare how each kernel responds to different geometric structures in the data.

## III. Results on Three Toy Datasets

### A. Linearly Separable Blobs

On the blobs dataset, the two classes form compact clusters on opposite sides of the feature space and are approximately linearly separable. An SVM with a linear kernel learns a straight decision boundary that passes between the clusters and achieves perfect test accuracy (1.00 in this run), as shown in Figure 1.

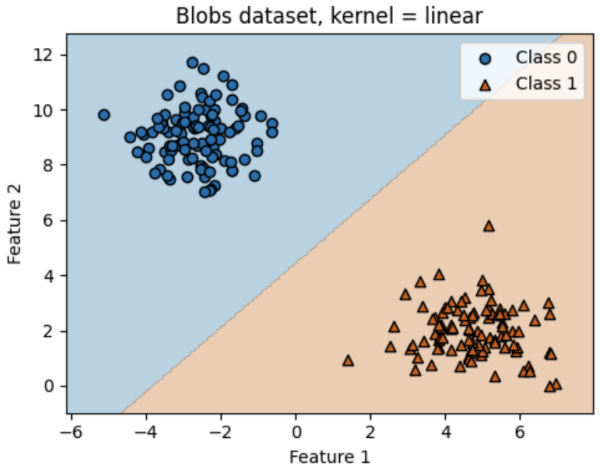


Figure 1: Decision boundary of an SVM with linear kernel on the blobs dataset. The boundary is a straight line that perfectly separates the two clusters (blue circles and orange triangles).

Test accuracy: 1.00.

The margin is clearly visible as a band between the two groups of points, illustrating the max-margin principle. Because the classes are already well separated by a straight line, there is no need for a more complex kernel. Using polynomial or RBF kernels would only introduce additional flexibility without improving generalisation. This experiment confirms that for data that are roughly linearly separable, a linear kernel is simple, interpretable, and highly effective[3][7][8].

### B. Moons Dataset

The moons dataset consists of two interleaving half-moon shapes and is not linearly separable in the original feature space[3][4]. With a linear kernel, the SVM can only draw a straight boundary. In my experiment this boundary cuts across both moons, misclassifying several points and achieving a test accuracy of about 0.83, as shown in Figure 2.

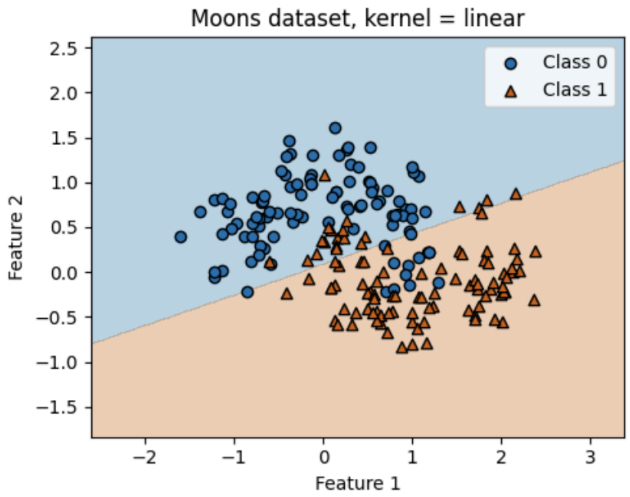


Figure 2: Decision boundary of an SVM with linear kernel on the moons dataset. The straight boundary misclassifies points where the two moons interleave. Test accuracy: 0.83.

When I switch to a polynomial kernel of degree 3, the decision boundary becomes curved and can follow the arc of the moons more closely (Figure 3). The boundary bends around one moon while separating it from the other, reducing the number of errors and increasing test accuracy to roughly 0.93.

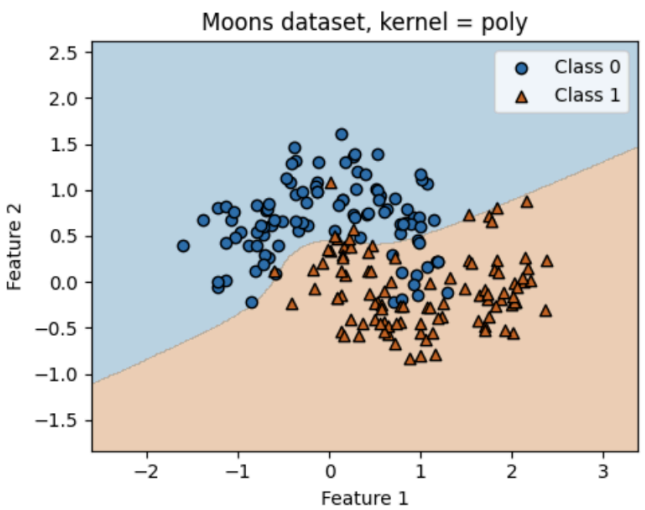


Figure 3: Decision boundary of an SVM with polynomial kernel (degree 3) on the moons dataset. The curved boundary better follows the moon shapes. Test accuracy: 0.93.

The RBF kernel yields a similar test accuracy (also around 0.93) but produces a boundary that adapts even more locally

to the data, smoothly wrapping around the moon shapes (Figure 4).

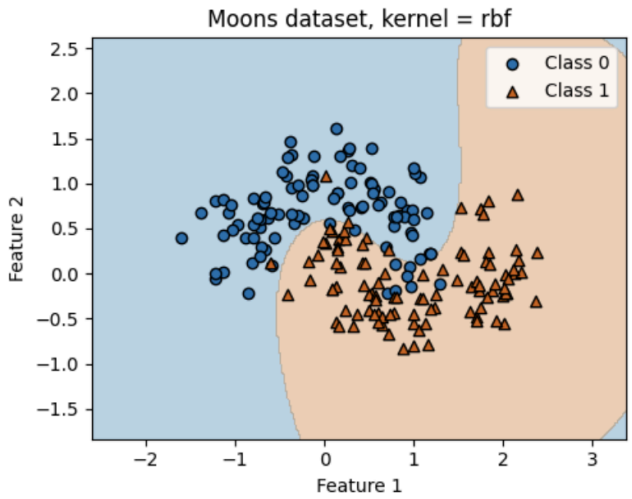


FIGURE 4: DECISION BOUNDARY OF AN SVM WITH RBF KERNEL ON THE MOONS DATASET. THE BOUNDARY SMOOTHLY CURVES AROUND THE TWO MOON SHAPES. TEST ACCURACY: 0.93.

These results show that non-linear kernels allow SVMs to capture curved decision boundaries in non-linearly separable data. Both polynomial and RBF kernels can model the moons reasonably well, while the linear kernel is too rigid[3][5][6].

### C. *Circles Dataset*

The circle's dataset contains two concentric rings: an inner circle and an outer circle[3][6]. No straight line can separate these classes, so we expect linear models to struggle. Indeed, an SVM with a linear kernel learns a vertical decision boundary that slices through both circles, resulting in many misclassifications and a low test accuracy of around 0.47 (Figure 5).

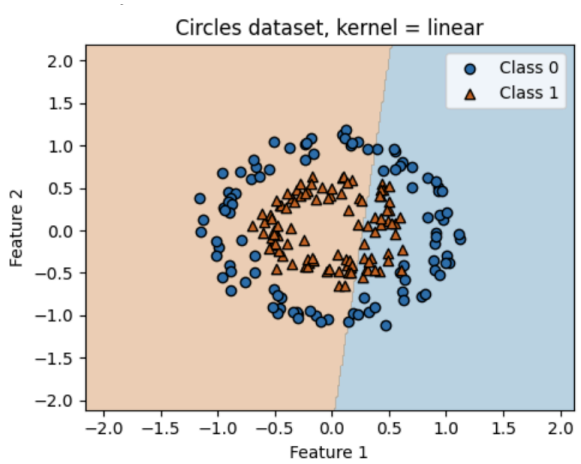


FIGURE 5: DECISION BOUNDARY OF AN SVM WITH LINEAR KERNEL ON THE CIRCLES DATASET. THE STRAIGHT VERTICAL BOUNDARY MISCLASSIFIES MANY POINTS FROM BOTH CIRCLES. TEST ACCURACY: 0.47.

Using a degree-3 polynomial kernel allows the boundary to bend, but in this configuration it still cannot form a clean closed curve around the inner circle. The decision region becomes warped yet remains roughly half-plane shaped, leading to only a modest improvement in accuracy (about 0.58), as shown in Figure 6.

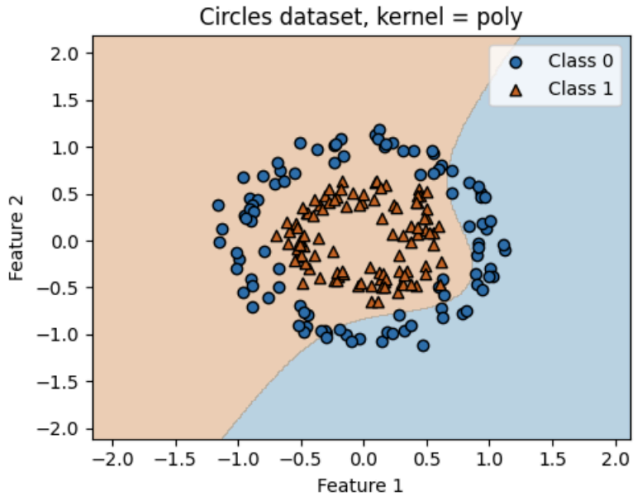


FIGURE 6: DECISION BOUNDARY OF AN SVM WITH POLYNOMIAL KERNEL (DEGREE 3) ON THE CIRCLES DATASET. THE BOUNDARY CURVES BUT CANNOT FORM A PROPER CLOSED SHAPE AROUND THE INNER CIRCLE. TEST ACCURACY: 0.58.

In contrast, the RBF kernel produces a circular decision boundary that tightly encloses the inner ring of points while excluding the outer ring (Figure 7). This matches the true structure of the data and yields a high test accuracy of about 0.98.

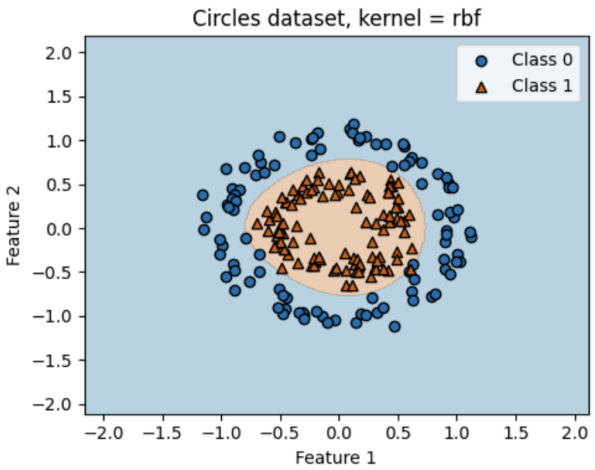


FIGURE 7: DECISION BOUNDARY OF AN SVM WITH RBF KERNEL ON THE CIRCLES DATASET. THE BOUNDARY FORMS A CIRCULAR SHAPE THAT SEPARATES THE INNER AND OUTER CIRCLES EFFECTIVELY. TEST ACCURACY: 0.98.

The circle's example highlights the strength of the RBF kernel for problems where classes form compact, nested clusters. By measuring similarity with a radial basis function, the RBF

kernel enables SVMs to draw smoothly curved boundaries that would be difficult to approximate with low-degree polynomial kernels[3][6][8].

### *D. Summary of Results*

Table I summarises the test accuracy for each kernel-dataset combination. On the blobs dataset, the linear kernel already reaches perfect accuracy, so more complex kernels are unnecessary[1][3]. For the non-linear moons and circles datasets, non-linear kernels (polynomials and especially RBF) consistently outperform the linear kernel[3][4][6]. This reflects the intuition that the kernel controls the shape of the decision boundary: linear kernels can only draw straight lines, while polynomial and RBF kernels can represent curved boundaries of varying complexity[1][3][11].

| Dataset | Kernel | Test Accuracy |
|---|---|---|
| **Blobs** | **Linear** | **1.000** |
| **Moons** | **Linear** | **0.833** |
| **Moons** | **Polynomial (deg=3)** | **0.933** |
| **Moons** | **RBF** | **0.933** |
| **Circles** | **Linear** | **0.467** |
| **Circles** | **Polynomial (deg=3)** | **0.583** |
| **Circles** | **RBF** | **0.983** |

TABLE 1: TEST ACCURACY OF SVM CLASSIFIERS WITH DIFFERENT KERNELS ON THREE DATASETS.

## IV. HOW TO CHOOSE A KERNEL IN PRACTICE

The experiments suggest several practical guidelines for kernel choice. When features are already informative and the separation between classes is roughly linear, a **linear kernel** is often sufficient[3][7]. Linear SVMs are computationally cheaper, involve fewer hyperparameters, and tend to generalise well when the boundary does not need to be highly curved.

When exploratory plots or prior knowledge indicate strong non-linear structure, a **non-linear kernel** is more appropriate. The **RBF kernel** is a common default because it can represent a wide range of smooth decision boundaries and has only two main hyperparameters ( and gamma) that govern the trade-off between margin size and boundary flexibility[1][11]. Larger values of gamma produce more complex, tightly curved boundaries, while smaller values yield smoother, simpler ones.

**Polynomial kernels** are useful when polynomial interactions between features are expected, but they can be more difficult to tune, since the degree, gamma, and all influence model complexity[1][5][11]. In many applications, practitioners start with a linear kernel, then try an RBF kernel if there is evidence of non-linear structure, and consider polynomial kernels only when there is a clear modelling motivation.

Regardless of the chosen kernel, cross-validation and careful hyperparameter search are essential to avoid overfitting and to estimate generalisation performance reliably[2][11].

## V. ETHICAL AND PRACTICAL CONSIDERATIONS

Kernel choice is not just a technical detail: it also affects how robust and reliable an SVM will be in real-world settings. Overly flexible kernels with poorly tuned hyperparameters can lead to **overfitting**, where the model captures noise rather than underlying patterns[11][12]. An overfitted classifier may perform very well on training data but behave unpredictably when deployed, especially if the data distribution shifts over time.

In high-stakes applications, such as healthcare, credit scoring, or predictive policing, unreliable models can amplify existing inequalities or make unjustified decisions[12][13]. Responsible development therefore requires more than maximising accuracy on a single test set. Practitioners should use validation procedures, monitor models after deployment, and where possible evaluate fairness metrics and robustness to distributional changes[13][14].

Transparency is also important: documenting which kernel was chosen, how hyperparameters were selected, and what limitations were observed helps others to audit and improve the model[13]. In teaching contexts, visualisations of decision boundaries, such as those in this tutorial, can make SVM behaviour more interpretable to non-experts, supporting informed use of the method.

## VI. KEY TAKEAWAYS

This tutorial used simple two-dimensional datasets to build intuition about how different SVM kernels shape decision boundaries. The main findings are:

- On linearly separable data (blobs), the linear kernel is simple and highly effective.
- On non-linear datasets (moons and circles), polynomials and especially RBF kernels can learn curved boundaries that match the data's geometry and deliver substantially better accuracy[3][6].

- In practice, kernel choice should balance model flexibility against the risk of overfitting, guided by exploratory analysis and validation results[11].
- Beyond pure performance, responsible use of SVMs also involves transparency, robustness checks, and attention to potential ethical impacts in real applications[12][13].

The accompanying Jupyter notebook demonstrates how these ideas translate into concrete code and visualisations that others can reuse and extend. All code and materials are available on

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